PROJECT REPORT

On

"Loan Approval Prediction System using Machine Learning"

Course – Fundamentals in AI and ML

Course Code – CSA2001

Submitted by:

RAKHI TYAGI

Reg No. 25BCE10780

Branch – B.Tech CSE

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VIT Bhopal University

Madhya Pradesh, India

1. Problem Statement and Reason for Choosing This Problem

In banks and financial institutions, deciding whether a loan should be approved or not is an important task. This decision depends on many factors such as income, credit history, number of dependents, and other personal details of the applicant. Usually, this process is done manually, which can take time and may also lead to inconsistent decisions.

I selected this problem because it is a real-life issue that affects both banks and people. Many students and individuals are familiar with the loan process, so it is easy to understand and relate to. It also helped me connect machine learning concepts with a practical application.

Another reason for choosing this problem is that it allowed me to learn the complete process of building a machine learning system, from data handling to creating a working application.

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2. Why This Problem Matters

This problem is important because loan approval decisions directly impact both financial institutions and applicants. Banks need to ensure that they give loans to people who can repay them, while applicants expect quick and fair decisions.

Manual processes can be slow and sometimes inaccurate. By using a machine learning system, the decision-making process can become faster and more consistent. It also reduces the chances of human error and helps in handling a large number of applications efficiently.

Overall, solving this problem shows how technology can be used to improve real-world systems and make them more efficient and reliable.

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3. Objectives of the Project

The main objectives of this project are:

- **To understand and apply machine learning concepts in a real problem-**The first objective is to learn how machine learning works in real-life situations. Through this

project, I focused on understanding how data is used to train a model and how predictions are made based on that data.

- **To build a loan approval prediction model-** Another important goal is to create a model that can predict whether a loan will be approved or not. This helps in understanding how classification problems are solved using algorithms like Logistic Regression.

- **To learn data preprocessing and data handling techniques-** This project also aims to improve skills in handling raw data, such as filling missing values, converting text into numbers, and preparing data before training the model.

- **To develop a simple and user-friendly web application**- I wanted to create an interface where users can easily enter their details and get results. For this, I used Streamlit to make the application interactive and easy to use.

- **To understand the complete workflow of a machine learning project-** The final objective is to learn the full process, starting from data understanding to model building and finally deploying it in a working application.

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4. Approach to Solving the Problem

To solve this problem, I followed a step-by-step approach in a proper sequence:

I. **First, I collected and studied the dataset** to understand what kind of data is available and what each column represents.

II. **Next, I cleaned and preprocessed the data** by handling missing values and fixing incorrect entries.

III. **Then, I converted categorical data into numerical form** so that the machine learning model could understand it properly.

IV. **After that, I selected Logistic Regression as the model** because it is simple and suitable for prediction problems.

V. **I then trained the model and checked its accuracy** using training and testing data.

VI. **After training, I saved the model** so it can be used again without retraining.

VII. **Next, I developed a web application using Streamlit** to make the system interactive.

VIII. **Finally, I connected the model with the application** to provide real-time predictions.

This approach helped me build the complete system step by step.

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5. Dataset Description

The dataset used in this project was taken from Kaggle. It contains information about different loan applicants along with their loan approval status. The dataset is useful for understanding how different factors affect the decision of loan approval.

The data includes both personal and financial details of the applicants. These details are used by the model to learn patterns and make predictions.

Features included:

- Gender
- Married
- Dependents
- Education
- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Term
- Credit History
- Property Area
- Loan Status (Target Variable)

The dataset helps the model understand past data and learn how different factors influence whether a loan is approved or not.

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6. Data Understanding

Before applying any machine learning model, I carefully explored and understood the dataset. This step helped me identify what changes were needed before training the model.

- Used **head( )** function to view the first few rows of the dataset

- Checked column names and understood what each feature represents
- Identified missing values in multiple columns
- Observed which data is categorical and which is numerical
- Noticed special values like “3+” in dependents column
- Checked unique values in columns to understand data variety
- Understood the importance of features like credit history and income

This step helped me clearly decide how to clean and preprocess the data for better model performance.

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7. Data Preprocessing

Data preprocessing was one of the most important parts of this project. It helped in preparing the raw dataset into a clean and usable format so that the machine learning model could work properly and give better results.

7.1 Handling Missing Values

In the dataset, some values were missing in different columns. Instead of removing those rows, I filled them carefully:

- Used **mode (most frequent value)** to fill missing values in categorical columns like Gender, Married, and Self Employed
- Used **mean (average value)** to fill missing values in numerical columns like Loan Amount

This ensured that no important data was removed and the dataset remained complete.

7.2 Data Cleaning

Some data values were not in the correct format, so cleaning was required:

- Converted “3+” dependents into numeric value 3
- Made sure all values were in a consistent format

This step was important to avoid errors during model training.

7.3 Encoding Categorical Data

Machine learning models cannot understand text data, so I converted categorical values into numbers:

- Male → 1, Female → 0
- Yes → 1, No → 0

- Property Area → Rural (0), Semiurban (1), Urban (2)

This step made the dataset suitable for training the model.

7.4 Feature Selection

Not all columns are useful for prediction. I selected only important features:

- Removed unnecessary column like **Loan_ID**
- Selected relevant features for model training

This helped in improving model performance and reducing complexity.

7.5 Data Preparation for Model

Before training the model, I prepared the dataset properly:

- Divided data into input features (X) and target variable (Y)
- Ensured all values were numerical
- Checked if any missing values were left

This final step made the dataset ready for training the machine learning model.

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8. Data Splitting

The dataset was divided into two parts:

- **Training Data (80%)**
- **Testing Data (20%)**

The training data was used to train the machine learning model, while the testing data was used to check how well the model performs on new and unseen data. This helps in understanding whether the model is learning correctly or just memorizing the data.

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9. Model Selection

I selected **Logistic Regression** for this project.

Reasons:

- **Suitable for binary classification**
  This problem has two outcomes (Approved or Not Approved), so Logistic Regression fits well for this type of task.

- **Easy to understand and implement**
  It is a simple algorithm, which helped me understand how the model works step by step.

- **Efficient for structured data**
  It performs well when working with numerical and structured datasets like this one.

10. Model Training and Evaluation

The model was trained using training data and tested using testing data.

- **Training accuracy was good**
  This shows that the model learned patterns from the training data properly.

- **Testing accuracy was around 77%**
  This indicates how well the model performs on new data.

This shows that the model performs reasonably well, but it is not perfect and can be improved further

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11. Key Decisions Made

During the project, I made several important decisions:

- **Chose Logistic Regression for simplicity**
  I selected Logistic Regression because it is easy to understand and suitable for this type of binary classification problem. It helped me clearly learn how the model works without making the project too complex.

- **Used mean/mode instead of removing data**
  Instead of deleting rows with missing values, I filled them using mean and mode. This decision helped in keeping the dataset complete and avoiding loss of useful information.

- **Added explanation system in the app**
  I decided to include an explanation section in the app so that users can understand why a loan is approved or not. This makes the system more user-friendly and practical.

- **Designed UI with help icons**
  I added help icons (ℹ️) for each input field so that users can easily understand what to enter. This improves the overall user experience.

- **Structured code with proper comments**
  I organized the code into sections and added comments to explain each part. This makes the code easy to read and shows my understanding of the project.

These decisions helped in improving both the usability of the application and my learning during the project.

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12. Challenges Faced

During development, I faced multiple challenges:

- **Handling missing values correctly**
  At the beginning, I was confused about whether to remove missing data or fill it. I learned to use mean and mode to handle missing values without losing important information.

- **Understanding encoding process**
  I found it difficult to understand how to convert text data into numbers. After practice, I understood how mapping works and why it is important for machine learning models.

- **Fixing Python and pip errors**
  While running the code, I faced errors like “pip not found” and module issues. I solved them by using correct commands and understanding how Python environment works.

- **Setting up environment in VS Code**
  Initially, I had issues with Python installation and interpreter selection. After setting the correct environment, everything started working properly.

- **Understanding model predictions vs real-world logic**
  Sometimes the model results did not match real-life expectations. I learned that models depend on dataset patterns and not real-world thinking.

- **Making explanation meaningful**
  It was challenging to give proper reasons for predictions. I improved it by adding simple logic and suggestions to make the output more understandable.

These challenges improved my problem-solving skills and helped me learn deeply.

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13. Web Application Development

A web application was developed using Streamlit. This helped in converting the machine learning model into a usable system where users can interact easily.

Features:

- **User-friendly interface**
  The app is simple and easy to use so that anyone can understand it without technical knowledge.

- **Input fields with help icons**
  Each input field has a small help description to guide the user about what to enter.

- **Clean UI design**
  Basic styling and proper layout were used to make the app look neat and organized.

- **Real-time prediction**
  The app gives instant results when the user clicks the predict button.

- **Probability display**
  It shows how confident the model is about its prediction.

- **Explanation and suggestions**
  The app also provides reasons and suggestions to help users understand the result better.

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14. Prediction System

The system works as follows:

- **User enters details**
  The user fills all the required fields like income, loan amount, and other details.

- **Data is converted into numerical format**
  All input values are converted into numbers so that the model can understand them.

- **Model predicts loan approval**
  The trained model processes the input and predicts whether the loan will be approved or not.

- **Output is displayed**
  The result is shown along with probability and explanation, making it easy for the user to understand.

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15. Explanation System

To improve usability, an explanation system was added so that users can understand the prediction result more clearly.

- **Provides reasons for rejection**
  It shows simple reasons when a loan is not approved.
- **Considers income, loan amount, credit history**
  It checks important factors that affect the decision.
- **Suggests improvements**
  It also gives basic suggestions to improve approval chances.

This makes the system more practical and easier to understand for users.

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16. Important Observations

- **Model predictions depend on dataset patterns**
  I observed that the model makes decisions based on the data it was trained on. It does not think like a human but follows patterns present in the dataset.
- **Sometimes results may not match real-world decisions**
  In some cases, the prediction did not match what we expect in real life. This happens because the model does not consider all real-world factors like loan purpose or job stability.
- **Probability helps understand confidence**
  The probability value helped in understanding how confident the model is about its prediction. A higher probability means the model is more sure about the result.
- **Explanation improves understanding**
  Adding explanation in the app made it easier to understand why a loan was approved or not. It helps users trust the system more.

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17. Limitations

- **Limited dataset features**
  The model is trained on a dataset with limited features, so it does not cover all possible factors that affect loan approval.
- **No loan purpose included**
  The dataset does not include the reason for taking the loan, which is an important factor in real-world decisions.
- **No job stability information**
  Details like job type or job stability are not present, which are usually considered by banks before approving loans.
- **Predictions may differ from real banking decisions**
  The model works based on data patterns, so its predictions may not always match real decisions taken by banks.

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18. Future Improvements

- **Add more features**
  In future, more real-world features like loan purpose, job stability, and credit score can be added to make the model more accurate.
- **Use advanced models**
  Instead of Logistic Regression, advanced models like Random Forest or XGBoost can be used to improve prediction performance.

- **Improve accuracy**
  By using better data and tuning the model, the accuracy of predictions can be increased.
- **Deploy online**
  The application can be hosted online so that users can access it from anywhere without running it locally.
- **Add login system**
  A login system can be added to make the application more secure and user-specific.

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19. What I Learned

Through this project, I learned many important concepts and practical skills:

- **Data preprocessing**
  I learned how to handle raw data, including filling missing values, cleaning incorrect data, and converting text into numerical form so that the model can understand it.
- **Model training**
  I understood how a machine learning model is trained using data and how it learns patterns to make predictions.
- **Model evaluation**
  I learned how to check model performance using accuracy and how to understand whether the model is working properly or not.
- **Streamlit app development**
  I learned how to create a simple web application where users can interact with the model and get results easily.
- **Code structuring and debugging**
  I improved my coding skills by organizing code into sections and solving errors during development.
- **Importance of documentation**
  I understood how important it is to explain the project clearly so that others can understand and use it easily.

This project helped me connect theory with practical implementation.

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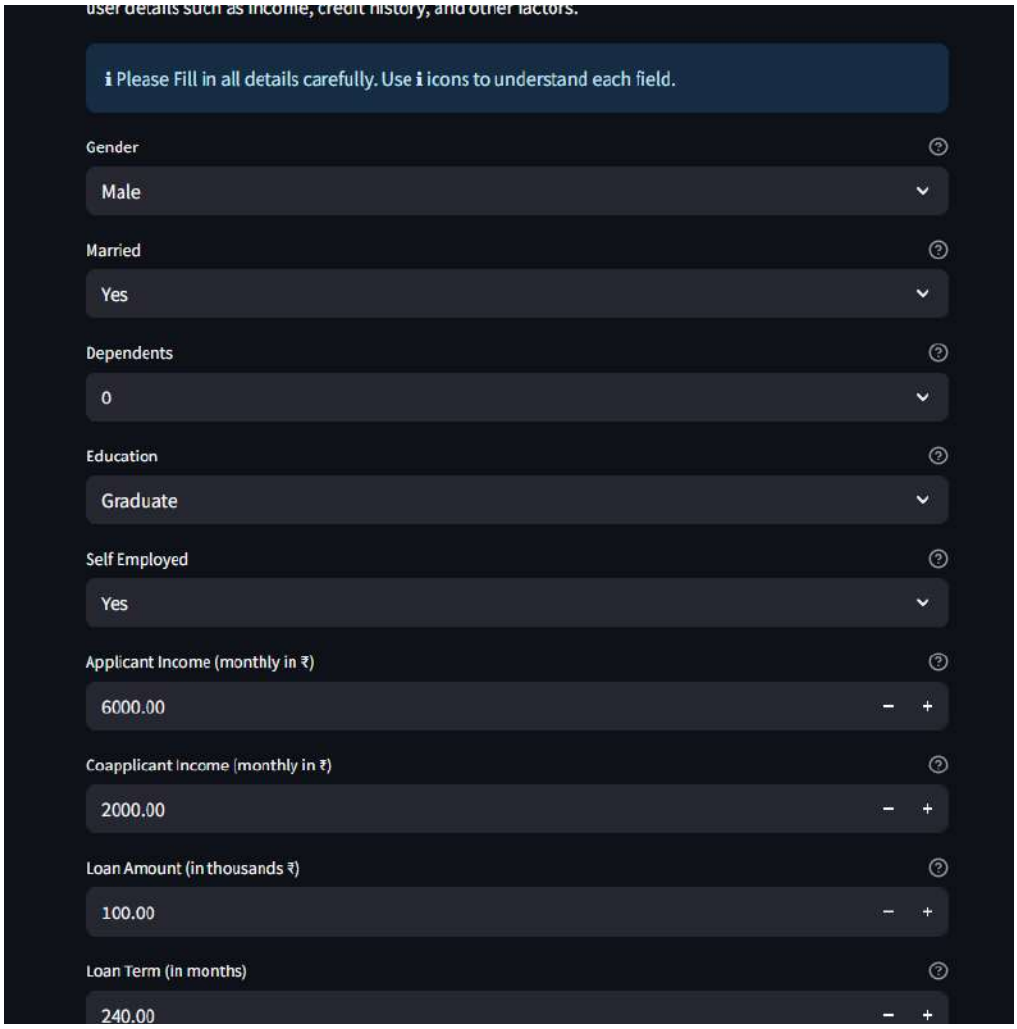
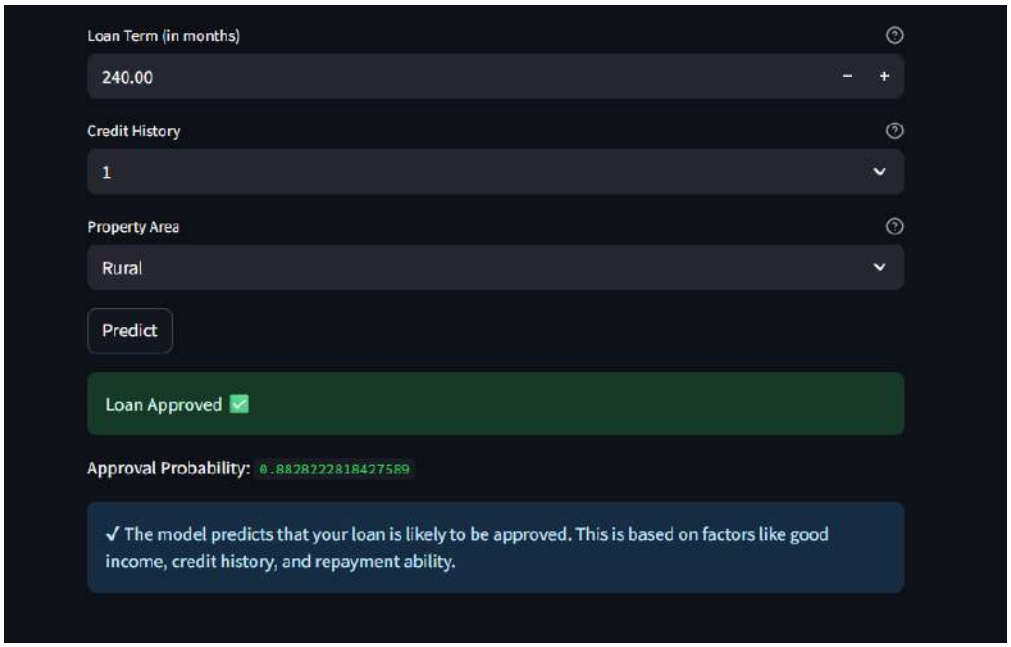
20. Model Testing and Results

To check the performance of the model, I tested it using different input values. I compared the **expected result (based on logic)** with the **actual result given by the model**. This helped me understand how the model behaves and where it may not match real-world thinking.

Sample Test Case 1 (Strong Profile)

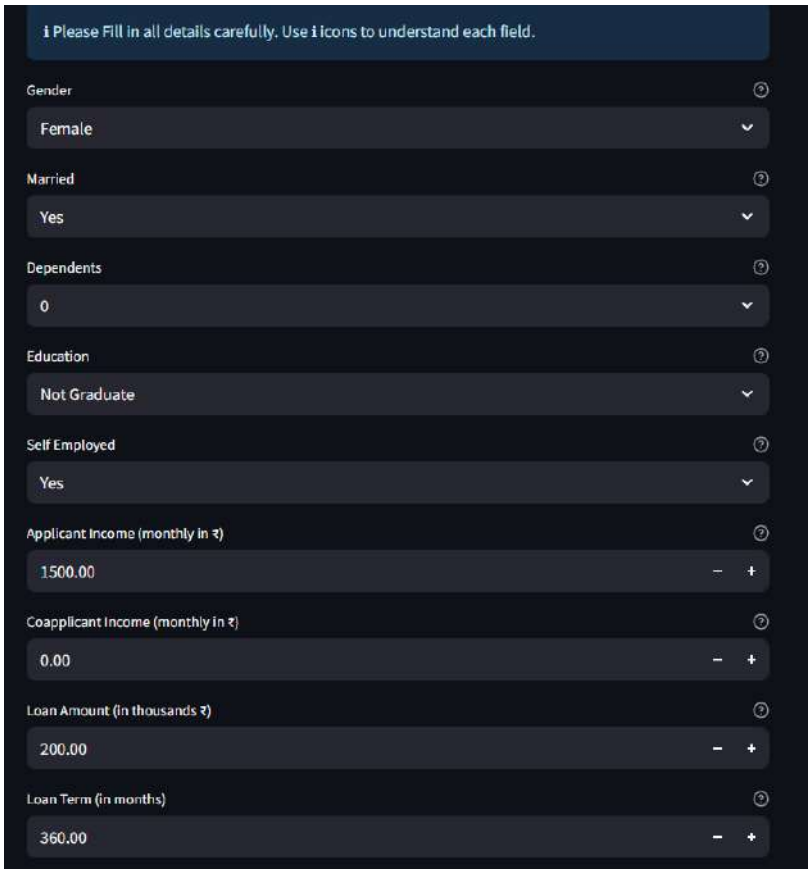
- **Input:**
  Income = 6000
  Coapplicant Income = 2000
  Loan Amount = 100
  Loan Term = 240
  Credit History = 1
  Dependents = 0
  Education = Graduate
- **Expected Output:** Loan Approved
- **Model Output:** loan approved

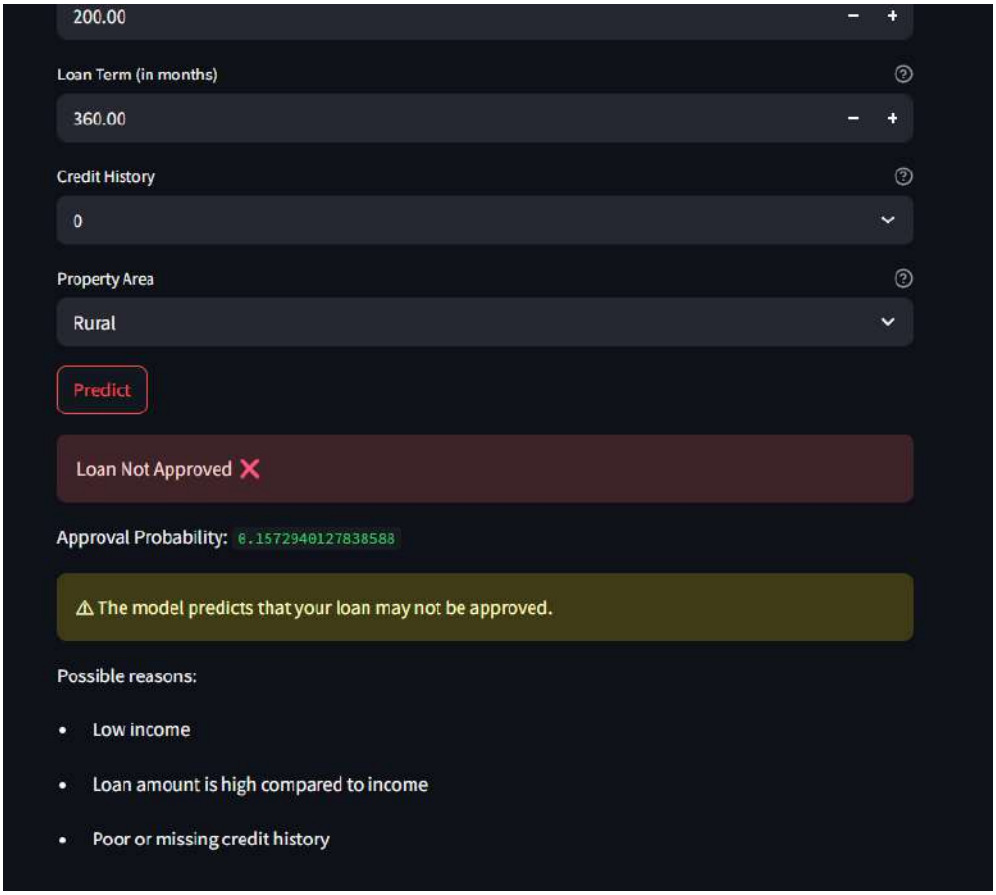




Sample Test Case 2 (Low Income + High Risk

- **Input:**
  Income = 1500
  Coapplicant Income = 0
  Loan Amount = 200
  Loan Term = 360
  Credit History = 0
  Dependents = 3+
  Education = Not Graduate
- **Expected Output:** Loan Not Approved
- **Model Output:** Loan Not Approved

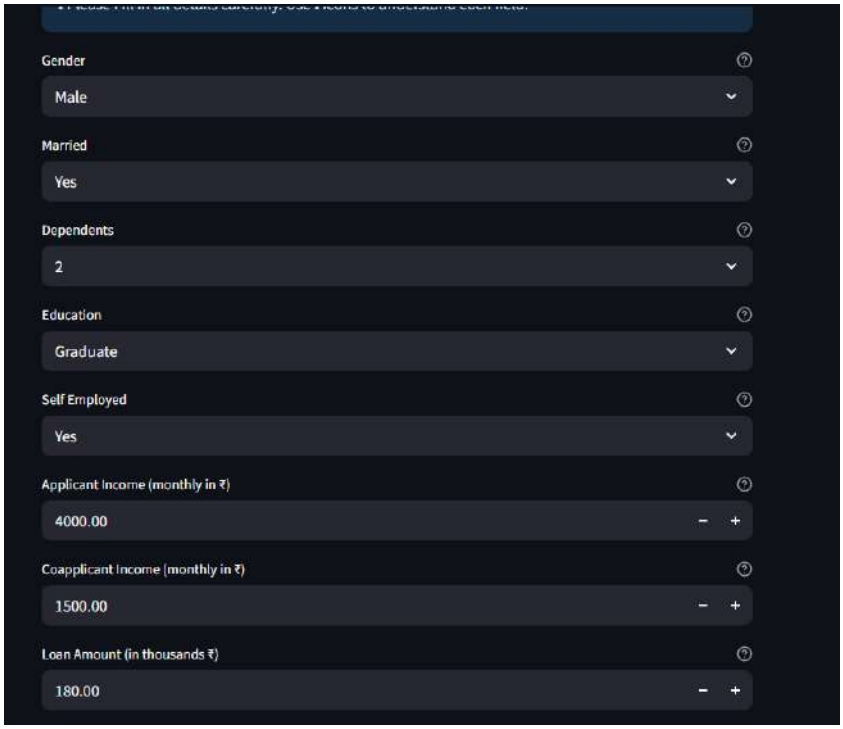


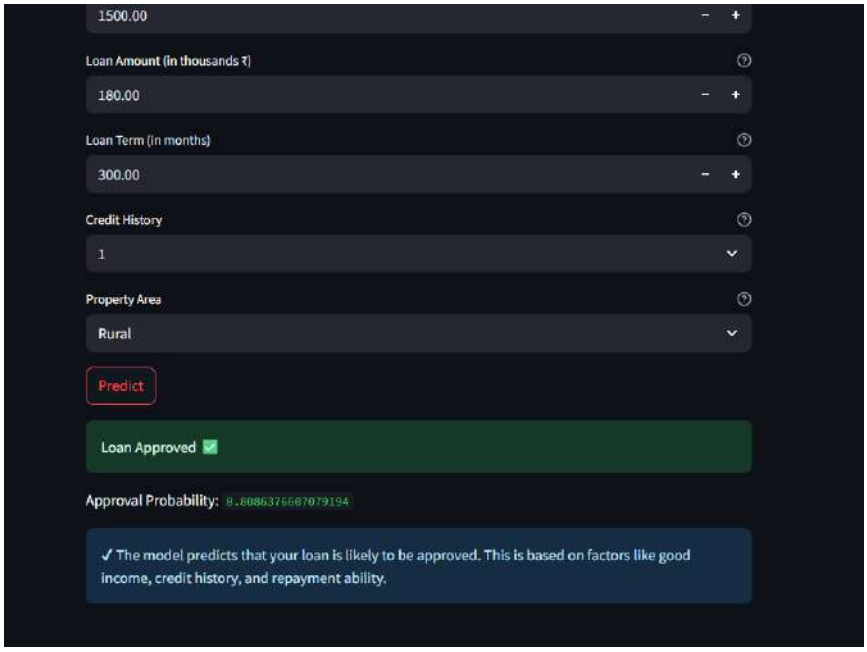


Sample Test Case 3 (Mixed Profile – Case)

- **Input:**
  Income = 4000
  Coapplicant Income = 1500
  Loan Amount = 180
  Loan Term = 300
  Credit History = 1
  Dependents = 2
  Education = Graduate

Expected Output: Loan Approved (based on good credit and moderate income)
Model Output: Loan Approved





In some cases, the model output may differ from the expected result because the model learns only from dataset patterns and does not consider all real-world factors.

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21. Conclusion

In this project, I successfully built a Loan Approval Prediction System using machine learning. I learned how to work with real-world data, clean it, and prepare it for model training. By using Logistic Regression, I was able to create a model that can predict whether a loan will be approved or not based on different input factors.

While working on this project, I understood that machine learning models depend completely on the data they are trained on. Sometimes, the predictions may not match real-life situations because the model does not consider all real-world conditions. This helped me understand the limitations of machine learning and the importance of good quality data.

Overall, this project helped me improve my practical knowledge and connect theory with real application. I also learned how to build a complete system from start to end, including a web application that users can interact with. This project was a valuable learning experience and gave me confidence in working on machine learning-based applications in the future.